XU Zengrui

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EDUCATION

Xi'anJiaotong University (Undergraduate) *2022.9-2026.7*

- School: XJTU-POLIMI Joint School of Design and Innovation
- **Major**: Industrial Design (MILANO), **Average Score:** 87.62; **Minor**: Computer Science and Technology， **Average Score:** 87
- Relevant Courses: Object-Oriented Programming (98), Data Structures and Algorithms (95), Computer Vision and Pattern Recognition (91), Introduction to Machine Learning (97), Advanced Mathematics-1(98), Advanced Mathematics-2(98), Linear Algebra (100), Probability and Mathematical Statistics (93)

RESEARCH & ACADEMIC PROJECTS

Summer Workshop: Deep Learning & Emotional Interaction Vehicle, The National University of Singapore (NUS), School of Computing (SOC) *2025.7*

- Developed a pedestrian detection and tracking system for an autonomous vehicle using the YOLOv8 + DeepSort model, enabling accurate real-time ID assignment and tracking of individuals in view.
- Built a facial emotion classifier(8 classes) based on YOLOv8, achieving mAP@0.5:0.95=0.497.
- Implemented vehicle state control logic through coding: Searching → Tracking → Emotion_Detect → Interaction.

Industrial Production Management System Based on Multi-AI Agent Collaboration *2025.3-2025.7*

- Fine-tuned the Qwen2.5-32B model on industrial corpora using parallel scheduling across four servers, enhancing the model's domain knowledge understanding and task-response capability.
- Developed an industrial scheduling RAG module with hierarchical strategies (chapter-level chunking + summary compression + title indexing) and prompt re-ranking, improving retrieval QA accuracy by +18 percentage points and reducing hallucination rate by –12 percentage points.
- Integrated MiniCPM multimodal model to enable visual extraction and cross-modal RAG QA on industrial images (e.g., scheduling charts, equipment diagrams).
- Built an MCP service combining production scheduling, knowledge QA, and image-text parsing; trained Function Call abilities for industrial tasks, achieving +3 percentage points on the BFCL benchmark.

Paper: 《Decoding Olympic Glory: A Data-Driven Approach to Medal Predictions and Strategic Insights》 (Published at MLIC, El-indexed conference) *2025.3-present*

- Expanded on our 2025 MCM award-winning project to build a more robust model and experimental framework.
- Introduced SHAP interpretability analysis to quantify feature impacts.
- Applied K-means clustering and correlation analysis to reveal strategic patterns in Olympic event selection by host countries.

Target Detection System for Wireless Car Charging Logos Based on Improved YOLOv11 *2025.3-2025.5*

- Integrated RepNCSPELAN4_high modules, MCA attention modules, and BiFPN modules into the YOLOv11 structure, improving mAP@0.5 from 0.891 to 0.937 and mAP@0.5:0.95 from 0.468 to 0.517.
- Applied various data augmentation techniques, such as Gaussian blur, random noise, random rotation, cropping, fog simulation, color jittering to enlarge the dataset and improve model robustness under limited training samples.
- Deployed the system to Raspberry Pi 5 for real-time recognition.

Age Estimation Model Based on Se_ResNeXt50_32x4d (Course Project: Computer Vision and Pattern Recognition) *2024.10-2024.12*

- Replaced simple baseline with Se_ResNeXt50_324d to enhance feature extraction via SE modules and residual connections, making MAE reduce from 7.1 to 5.5 and training loss from 3.3 to 2.0.

Multi-Objective Learning Optimization for Fine Ranking under Dual-Column Recommendation *2024.9.2024.12*

- Applied a MMoE (Multi-gate Mixture-of-Experts) model for joint training, achieving baseline AUCs of 0.9218 (Click), 0.9120 (Like), 0.9201 (Favorite), 0.9218 (Comment).
- Refined experts into shared and task-specific experts, and integrated TRM modules (from DTRN) to mitigate negative transfer, improving AUCs compared to the baseline (+0.08% Click, +0.16% Like, +0.18% Favorite, +0.07% Comment).
- Incorporated ESMM-style objectives (Click→Like/Comment/Favorite) with shared feature tables to mitigate bias and sparsity, delivering additional improvements (+0.20% Click, +0.17% Like, +0.21% Favorite, +0.13% Comment).
- Adopted Uncertainty Weight for multi-objective loss fusion, outperforming equal-weight fusion with gains of (+0.04% Click, +0.06% Like, +0.02% Favorite, +0.08% Comment).

"Abyss Hunter": Autonomous Underwater Vehicle Design Based on YOLO *2023.11-2024.3*

- Performed data augmentation and dual-path enhancement on marine image datasets: one path using red compensation and gamma correction, the other using blue compensation and sharpening.
- Fused enhanced images using Laplacian and Gaussian pyramids to improve brightness, color fidelity, and detail clarity, resulting in improved YOLOv8 detection performance.
- Won First Prize in Innovation Track, 35th "Tengfei Cup" Innovation & Entrepreneurship Competition, Xi'an Jiaotong University.

COMPETITION EXPERIENCE

Tencent Advertising Algorithm Competition *2025.8-2025.9*

- Replace BCE Loss(Binary CrossEntropy Loss) with InfoNCE Loss (+0.015).
- Integrated multiple temporal features such as interaction interval, time decay into the Embedding Layer (+0.003).
- Reconstructed the model's attention module from Self-attention blocks to HSTU (+0.004).
- Designed and implemented a three-tower feature fusion architecture based on an improved SENet (including skip connection and attention pooling), replacing the original DNN dual-tower structure (+0.007).
- Achieved a score of 0.0708489, ranking 350/1334.

Kaggle Competition-CMI: Detect Behavior with Sensor Data *2025.8-2025.9*

- Built a unified multi-model ensemble pipeline including CNN, GRU, Bi-LSTM, and Transformer to process all data types: IMU, THM, TOF.
- Designed and implemented a dual-branch prediction structure: one branch for IMU-only data (50% of dataset), the other for multi-modal data, improving robustness and accuracy.
- Achieved a score of 0.825027 on the private leaderboard, ranking 230/2657, awarded Bronze Medal.

Mathematical Contest in Modeling (MCM), COMAP (USA) *2025.1*

- Team member responsible for modeling and coding; Developed XG-Prophet hybrid model for Olympic medal prediction (MAEs: Gold 1.09, Silver 0.95, Bronze 1.12, Total 2.24).
- Built GRU-ARIMA+XGBoost ensemble model for first-time winner prediction (AUC: 0.917).
- Applied Difference-in-Differences and event study to assess the "Great Coach Effect".
- Awarded Honorable Mention (H Prize)

CAMPUS ACTIVITIES & HONORS

- National Undergraduate Mathematics Competition, Shaanxi Region (Provincial First Prize) *2024.11*
- Recipient of Second-Class Scholarship, Xi'an Jiaotong University (University-level) *2023.10*

STANDARDIZED TESTS

- IELTS: Overall 6.5 (W: 6.0, R: 7.5, L: 6.0, S: 5.5), 2025.4.26
- CET-4 (College English Test Band 4): 613, 2023.6.17

SKILLS

- Programming Languages: Python, Java, C++, C, JavaScript
- Techniques: Pytorch, TensorFlow, React, Spring Boot